Lab 6.1 - Snapshot, Fault, Restore

KVM Virtualisation on Oracle Linux - Day 2, Module 6

Overview

In this lab you will take an internal snapshot of ol-lab-01 while it's running, deliberately break something that would normally require real troubleshooting to fix, and then restore the VM to its pre-fault state with a single command. The fault used here - disabling remote SSH access - is a deliberately realistic one: the kind of self-inflicted lockout every administrator eventually causes at least once.

This lab assumes ol-lab-01 exists, is running, and is reachable over SSH on its bridged network address from Lab 4.1.

Learning Objective

By the end of this lab, you will be able to:

- Take an internal snapshot of a running VM covering both disk and memory state
- Confirm a fault has actually occurred before attempting a fix
- Revert a VM to a snapshot and verify the fault is gone
- Inspect a VM's snapshot history with snapshot-list --tree

Step by Step Guide

Step 1: Take a Snapshot in a Known-Good State

Before touching anything, confirm the VM is running and reachable, then capture its current state.

1. Confirm the VM is running and note its address:

```
virsh list --all
virsh domifaddr ol-lab-01
```

2. Confirm SSH access works right now, from the host:

```
ssh oracle@<ol-lab-01's address> 'echo SSH is working'
```

Expected result: *SSH is working printed back - confirms remote access is functional before you break anything.*

3. Take the snapshot:

```
virsh snapshot-create-as ol-lab-01 pre-fault-snapshot \
  "Known-good state, SSH working"
```

Expected result: *Domain snapshot pre-fault-snapshot created.*

4. Confirm it exists:

```
virsh snapshot-list ol-lab-01
```

Expected result: *pre-fault-snapshot listed, State running - since the VM was running and no --disk-only flag was given, this snapshot includes memory as well as disk.*

Step 2: Deliberately Break the Guest

Simulate a common self-inflicted fault: disabling SSH access on the VM itself. Do this from the VM's console (via virt-viewer), not over the SSH session you're about to cut off.

5. Connect to the console:

```
virt-viewer --connect qemu:///system ol-lab-01
```

6. Inside the guest, stop and disable the SSH service:

```
sudo systemctl stop sshd
sudo systemctl disable sshd
```

Expected result: *Both commands complete with no error. sshd is now neither running nor set to start on boot.*

Note: *This fault is deliberately chosen because it only affects remote access, not the console - you can still reach the VM to fix or revert it, exactly as you could on a real system if you had physical or hypervisor console access.*

Step 3: Confirm the Fault

From the host, confirm the fault is real before attempting any fix - don't assume.

7. Attempt to SSH in again:

```
ssh oracle@<ol-lab-01's address> 'echo SSH is working'
```

Expected result: *Connection refused or a timeout - SSH is no longer reachable, confirming the fault.*

8. Confirm from the console that the service is indeed the cause:

```
sudo systemctl status sshd
```

Expected result: *Shows inactive (dead), confirming the service is stopped.*

Step 4: Revert to the Pre-Fault Snapshot

9. Revert the VM:

```
virsh snapshot-revert ol-lab-01 pre-fault-snapshot
```

Expected result: *Command completes with no error. Because the snapshot included memory state, the VM resumes running immediately - no separate virsh start needed.*

10. Give the VM a few seconds to settle, then confirm SSH access is restored:

```
ssh oracle@<ol-lab-01's address> 'echo SSH is working'
```

Expected result: *SSH is working printed back - the fault is gone, exactly as if it had never happened.*

11. Confirm from the console that sshd is active again:

```
sudo systemctl status sshd
```

Expected result: *Shows active (running) - the revert restored the service to the state it was in at snapshot time, without you needing to manually re-enable anything.*

Step 5: Review the Snapshot

12. List the snapshot in tree form:

```
virsh snapshot-list ol-lab-01 --tree
```

Expected result: *pre-fault-snapshot listed as the only entry, with no parent - since this is the first and only snapshot taken.*

13. Inspect its details:

```
virsh snapshot-info ol-lab-01 pre-fault-snapshot
```

Expected result: *Shows Current: yes (it's the most recently reverted-to snapshot), State: running, and Location: internal - confirming this was an internal, disk-and-memory snapshot the whole way through.*

Note: This snapshot is still sitting on ol-lab-01's disk, taking up space. In a real environment, once you've confirmed the VM is stable, it's good practice to delete a rollback snapshot you no longer need with virsh snapshot-delete ol-lab-01 pre-fault-snapshot - left for this lab so you can review it in Step 5.

Outcomes

By the end of this lab, you have taken a live internal snapshot of ol-lab-01, deliberately broken remote access to it, confirmed the fault was real, and reverted the VM back to a fully working state with a single command - no manual troubleshooting of the sshd service required. This is the exact safety net that makes it reasonable to attempt risky changes on a VM in the first place, which is precisely what the remaining modules today will keep doing.